

The Institutionalization of Money: Advancing Fundamental Institutional Theory Through Inverse Generative Social Science

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Extended Abstract

Generative Social Science seeks to understand complex socioeconomic systems by simulating micro-level agent behavior and observing emergent macro-level patterns. In contrast, *inverse* Generative Social Science (iGSS) begins with empirically observed macro-level patterns and, through computational optimization techniques like evolutionary algorithms, searches for plausible micro-level rule sets that could have produced them. Although iGSS has mostly been applied to narrowly defined empirical problems, it holds a broader promise to illuminate foundational questions of the social and economic sciences, including the problem of microfoundations in economics, the role of rules and equilibria in the formation of institutions, and the drivers of different forms of socioeconomic organization.

This paper revisits a well-established tradition of "fundamental models of money" in pure economic theory, examines major critiques raised against these abstract formal models by monetary scholarship in history and sociology, and deploys the theory-agnostic toolset of iGSS to develop a synthetic agent-based framework bridging the gap between these perspectives.

Money is a centerpiece of the socioeconomic world, a key aspect of the micro-macro link in economics, and arguably the core social institution enabling all other forms of economic organization. Among the many competing views of the origin of money, a common thread is the understanding that economies in which the use of money is widespread support higher levels of division of labor—that is, of economic complexity—than those relying on more direct interpersonal relations.

We develop an agent-based model posing division of labor problems that can only be solved if agents learn to coordinate and adopt mutually consistent decision rules for the distribution of goods among them. Agents are spatially separated, different agents produce different goods, and all agents benefit from the consumption of all goods. Agents can recognize and interact with each other, and transfer of goods is possible, but limited by proximity and time constraints. At this level of abstraction, our setup is compatible with both well-established fundamental models of economic systems and stylized observations from economic history, making no strong assumptions favoring any of these perspectives.

We then apply an iGSS framework to this setup. We define a "high level of division of labor" (i.e., the widespread consumption of a diverse set of goods by many agents) as a macro-level target and implement evolutionary algorithms at the agent level to discover decision rulesets for goods distribution. Simulations run until the system converges on a sufficiently high level of division of labor. We then evaluate which, among the emergent rules discovered by the model, might correspond to foundational economic institutions such as money.

We test this framework in three different scenarios. The first, "methodologically individualist iGSS," deploys the evolutionary discovery inverse-generative framework as it has been applied in most applications to date. We show that this is insufficient to generate decision rules that are complex and heterogeneous enough to be characterized as an economic institution. This partially confirms previous results from the economics literature. In two additional scenarios, we incorporate different "sociological primitives" into agent specifications. In the "institutional iGSS – exchange" scenario, we predefine an elementary spot exchange mechanism allowing agents to swap goods. In the "institutional iGSS – debt" scenario, agents are instead endowed with the ability to issue a personal IOU in exchange for receiving a good. We also explore a fourth scenario where decision rulesets are evolved at the level of links between pairs of agents, delineating an important direction for future research.

Applying the iGSS framework to agents with complex relational behavior greatly expands the space of possible rulesets for goods distribution, shedding new light on processes of economic self-organization. We find that both exchange and debt can serve as scaffolding for the evolution of increasingly complex rules, which, in turn, lead to an increasingly effective division of labor. We interpret these shared rulesets as meso-level institutions and argue that some of them can be understood as a virtual, emergent version of the institution of money. By placing institutions as the outcome of the modeling process in a theory-agnostic framework, this work provides a new, direct connection between abstract models and fundamental sociological theory. It also showcases the potential of iGSS as a general epistemological approach for realistic and empirically informed theory development in the social and economic sciences.

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Model Target	High levels of product consumption by agents (proof-of-concept) Theoretical or empirical monetary patterns (ideally)
Agent	• Location and time* • Bounded-length inventory of goods (replenished automatically based on location) • Bounded-length list of other agent IDs • Auxiliary type tags
Primitives	• At least one global macro-state variable • <i>Interpersonal operations:</i> “spot exchange” “interpersonal debt”
Combinators	Topology: agents fixed in space, interaction within range (translatable into random graph with differentiated agents) if-then =<> not and or + - * / “has” (inventory function; necessary for interpersonal interaction) mutate, crossover
Fitness Metric	Number of different items consumed by all agents per time step
Evolutionary Program	<i>Evolutionary Model Discovery</i>
Stopping Rule	Loss value threshold

Table 1: iGSS model setup overview